

Product Carbon Footprint Study Report

Product Name: ammonium sulfate production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: ammonium sulfate production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg ammonium sulfate

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg ammonium sulfate.

2. System Boundary

ammonium sulfate production, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of ammonium sulfate production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1.629 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

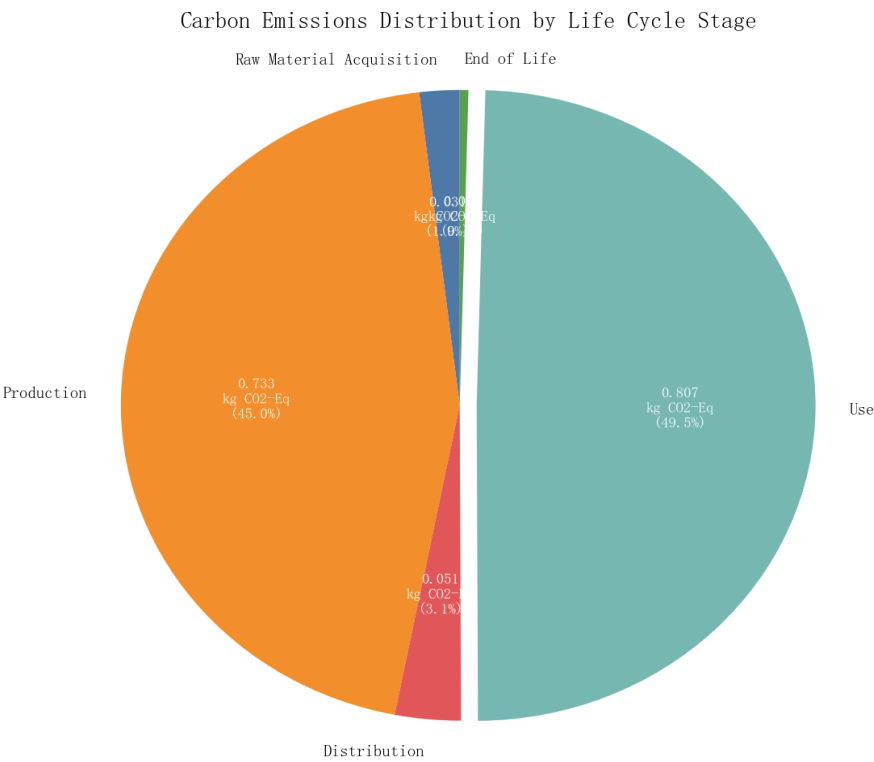


Table 1: Carbon Emissions by Life Cycle Stage

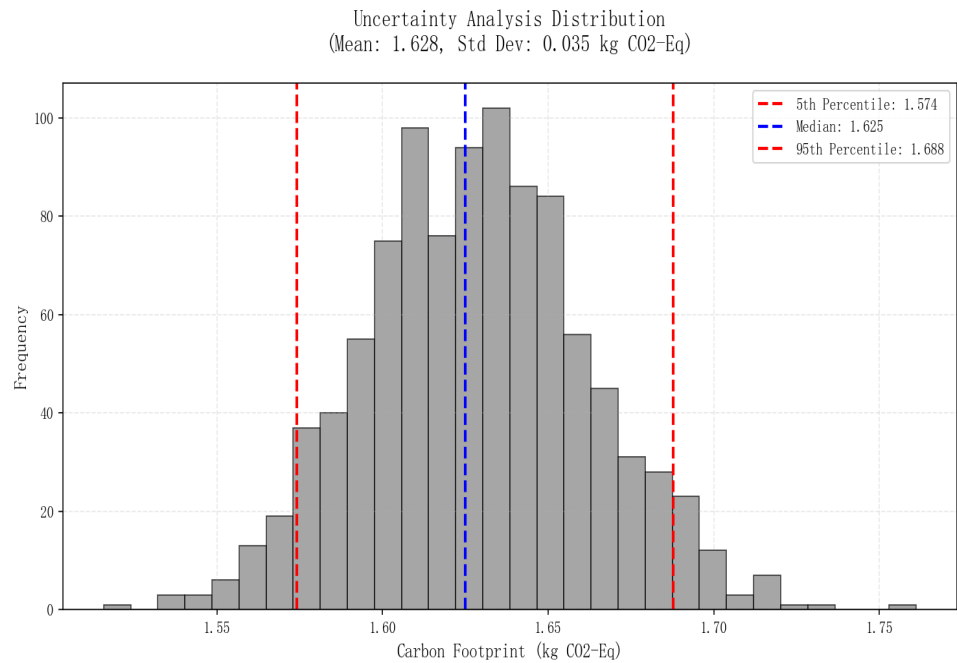
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.031	1.91
Production	0.733	45.01
Distribution	0.051	3.14

Use	0.807	49.53
End of Life	0.007	0.41
Total	1.629	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	1.628
Standard Deviation	0.035
Coefficient of Variation (CV)	2.12%
Median	1.625
5th Percentile	1.574
95th Percentile	1.688

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of ammonium sulfate production stands at 1.629 kg CO₂-Eq, representing a relatively contained emission intensity for this industrial chemical. The lifecycle assessment reveals a clear bimodal distribution of impacts, with the use phase accounting for 49.5% (0.807 kg CO₂-Eq) and the production phase contributing 45.0% (0.733 kg CO₂-Eq). These two stages together dominate nearly 95% of the overall footprint, while raw material extraction (1.9%), distribution (3.1%), and end-of-life treatment (0.4%) remain marginal.

This pattern indicates that emissions arise primarily from energy-intensive manufacturing operations and downstream application processes rather than material sourcing or logistics. The use-phase dominance suggests potential releases linked to fertilizer deployment or associated chemical reactions, whereas production impacts likely stem from ammonia synthesis and sulfuric acid neutralization steps. Identifying these hotspots enables targeted interventions to achieve meaningful decarbonization without addressing the entire value chain uniformly.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the selection of a cradle-to-grave system boundary with a functional unit of one kilogram of ammonium sulfate, reliance on generic process data from established LCA databases, and allocation of multi-output processes based on economic value. The analysis presumes average European energy mixes and standard agricultural application rates during the use phase, while excluding indirect land-use changes or biogenic carbon uptake.

Limitations arise from the use of secondary data sources that may not fully capture site-specific technologies or regional variations in raw material quality and efficiency. Data quality is further constrained by potential temporal mismatches and incomplete coverage of minor co-products or trace emissions. System boundary restrictions exclude upstream infrastructure construction, consumer behavior variability, and long-term soil carbon dynamics, which could alter the relative importance of the use phase under different scenarios.

3. Suggestions for Improvement

- Optimize production energy efficiency by transitioning to renewable electricity sources and waste-heat recovery systems to reduce the 0.733 kg CO₂-Eq manufacturing burden.
- Develop controlled-release formulations or precision application technologies in the use phase to lower the 0.807 kg CO₂-Eq emissions associated with fertilizer deployment.
- Conduct primary data collection at the production facility to replace generic database values and improve accuracy of the dominant stages.
- Explore process integration with carbon capture during ammonia synthesis to achieve measurable reductions in the production hotspot.
- Perform sensitivity analysis on use-phase assumptions to validate emission factors and identify region-specific mitigation opportunities.

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